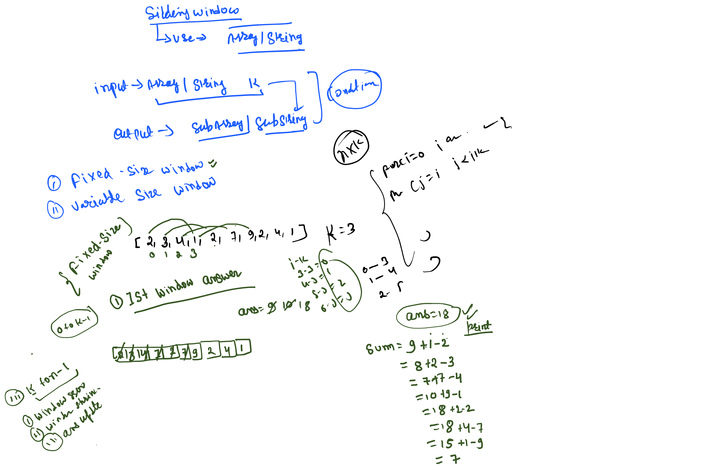
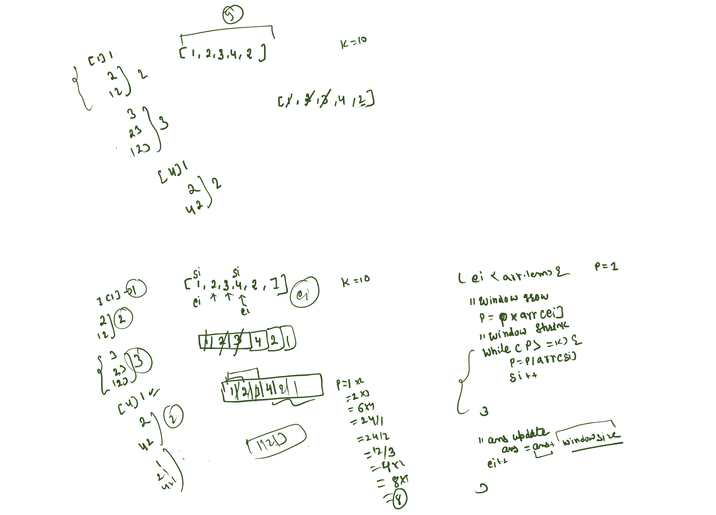
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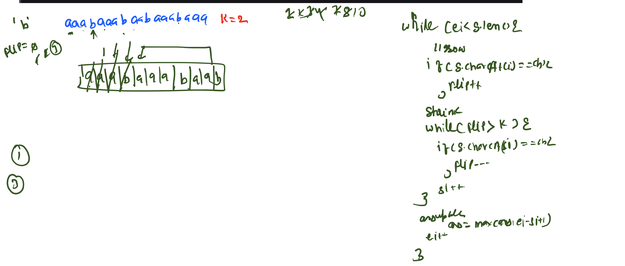



```
public static int Product_Less_Than_K(int[] nums, int k) {
        int si = 0, ei = 0, p = 1, ans = 0;
        while (ei < nums.length) {
            // window grow
            p = p * nums[ei];
            // window shrink
            while (p >= k) {
                p = p / nums[si];
                si++;
            }
            // ans update
            ans = ans + (ei - si + 1);

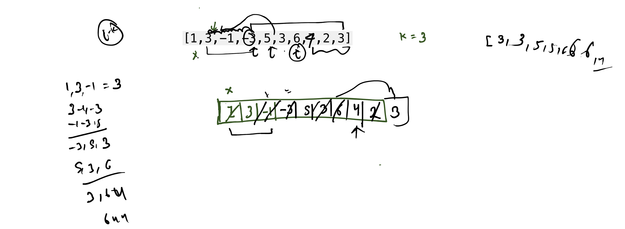
            ei++;
        }
        return ans;
    }
```

Kartik Bhaiya And Strings

Kartik Bhaiya has a string consisting of only 'a' and 'b' as the characters. Kartik Bhaiya describes perfectness of a string as the maximum length substring of equal characters. Kartik Bhaiya is given a number **k** which denotes the maximum number of characters he can change. Find the maximum perfectness he can generate by changing no more than **k** characters.




```
nums = [1,3,-1,-3,5,3,6,7], k = 3
```




```
public static int[] Window_Maximum(int[] arr, int k) {
        Deque<Integer> dq = new LinkedList<>();
        int[] ans = new int[arr.length - k + 1];
        int j = 0;
        for (int i = 0; i < k; i++) {
            while (!dq.isEmpty() && arr[i] > arr[dq.getLast()]) {
                dq.removeLast();
            }
            dq.add(i);
        }
        ans[j++] = arr[dq.getFirst()];
        for (int i = k; i < arr.length; i++) {
            // grow
            while (!dq.isEmpty() && arr[i] > arr[dq.getLast()]) {
                dq.removeLast();
            }
            dq.add(i);
            // shrink
            if (dq.getFirst() == i - k) {
                dq.removeFirst();
            }
            ans[j++] = arr[dq.getFirst()];
        }
        return ans;
    }
```

